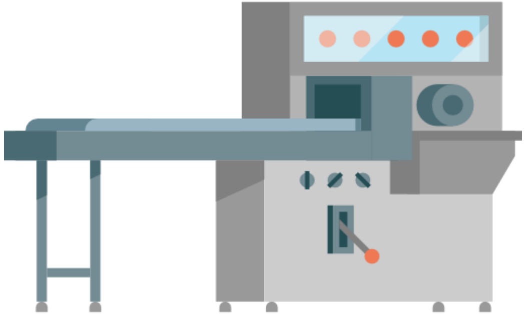
Technische Hochschule Rosenheim

MECHANICAL DESIGN

THE DESIGN OF A SMART TEACH-IN STATION

08/07/2024

Guided By: Dr. Ihab Ragai



Group-D

Team members:

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- Aniket Barotkar- 1066937
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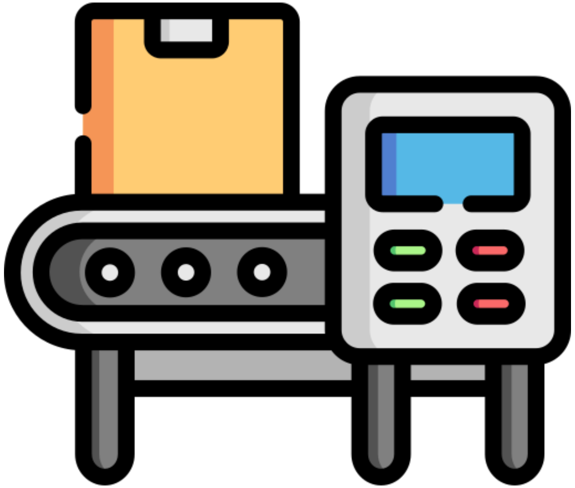
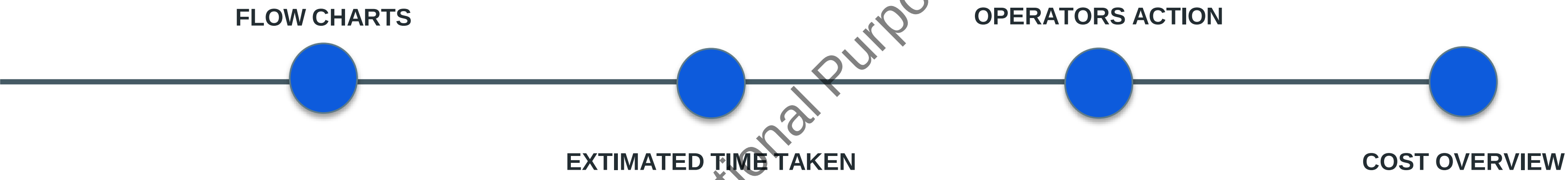


TIMELINE

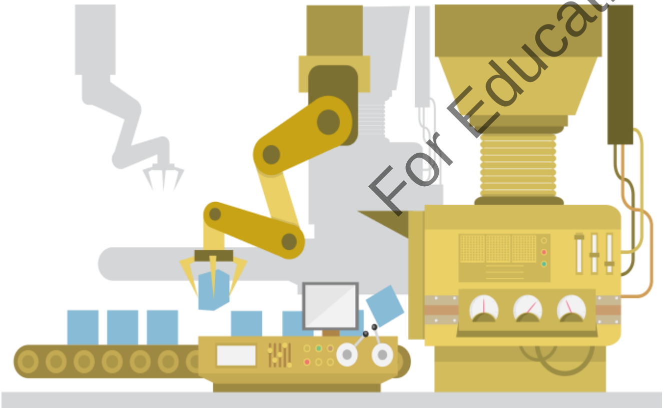
ABOUT PRESENTATION

- PROBLEM STATEMENT
- OBJECTIVE
- DESCRIPTION OF DIFFERENT CONCEPTS
- EXPLANATION OF MAIN DESIGN


FLOW CHARTS
EXTIMATED TIME TAKEN
OPERATORS ACTION
COST OVERVIEW

PROBLEM STATEMENT

- The absence of an effective solution for stacking different categories of products.
- Designing a teach-In station for all products
- Avoid instabilities during transport and handling.
- Solid products to bottom, less stable products on top.
- Features of each product is to be measured.



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OBJECTIVE

1. Identify the most suitable characteristic properties to measure the products:
   - Ensure rigidity
   - Detect defects (uglies)
   - Assess stackability
   - Facilitate product identification

2. The station is configurable as standalone device or as inline device

3. Product dimension range: from 120mm x 80mm x 40mm to 600mm x 400mm x 450mm

4. Product weight range: 0.5kg to 30kg

5. Operating speed: from 20 to 60 products per hour

BRIEF INTRODUCTION ABOUT DIFFERENT CONCEPTS

CONCEPT-1

Key Components: Utilized a 3D camera for measuring dimensions, calculating stiffness, and object identification.
Benefits: High precision in dimension and stiffness measurements, effective object identification.
Limitations: Complexity and potential high cost of whole system.

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CONCEPT-2

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Benefits: Accurate internal structure analysis.
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CONCEPT-3

Key Components: Thermal imaging for liquid detection and stiffness measurement.
Benefits: Non-contact measurement method, useful for detecting liquid presence.
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CONCEPT-4

Key Components: Similar to Concept 2 with an X-ray machine but incorporated different secondary components.
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Limitations: Shared the high cost issue with Concept 2, leading to its rejection.

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CONCEPT-4	**Key Components**: Similar to Concept 2 with an X-ray machine but incorporated different secondary components. **Benefits**: Maintained benefits of X-ray precision. **Limitations**: Shared the high cost issue with Concept 2, leading to its rejection.
CONCEPT-5	**Key Components**: Machine vision system for stiffness calculations and product identification (new vs. regular). **Benefits**: Effective for identifying product history and stiffness. **Limitations**: Potential issues with the accuracy and complexity of machine vision algorithms.

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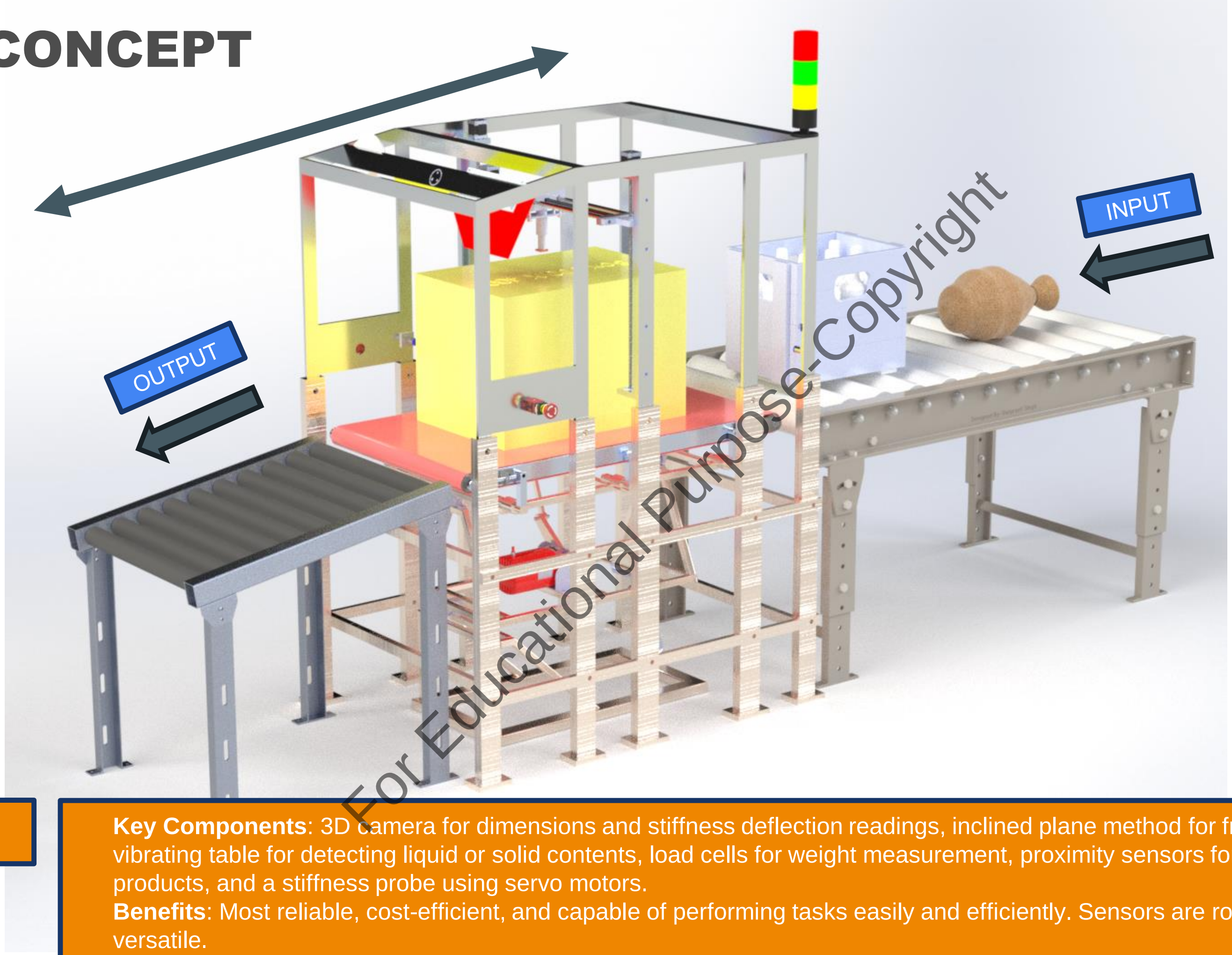
CONCEPT-6

Key Components: Focused on devices like 3D scanners, RFID tags, and optical measurement devices.
Benefits: Leveraging existing infrastructure.
Limitations: Limited versatility and multiple complex system which result in high cost.

FINAL CONCEPT



CONCEPT-7

Key Components: 3D camera for dimensions and stiffness deflection readings, inclined plane method for friction testing, vibrating table for detecting liquid or solid contents, load cells for weight measurement, proximity sensors for counting products, and a stiffness probe using servo motors.
Benefits: Most reliable, cost-efficient, and capable of performing tasks easily and efficiently. Sensors are robust and versatile.
Limitations: Complex integration but outweighed by overall performance.

PHYSICAL SIZES

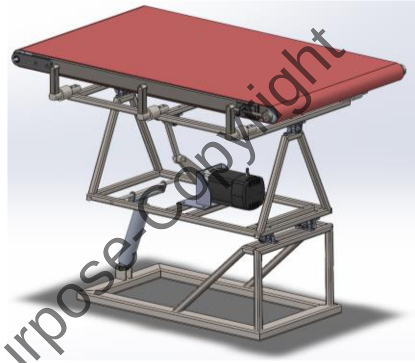
1.Table Dimensions:
- 1000 mm (Length) × 650 mm (Width) × 600 mm (Height)

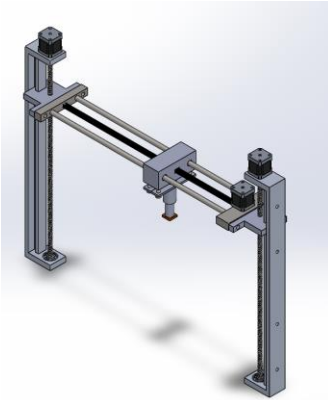
2.Stiffness Assembly Dimensions:
- 150 mm (Length) × 780 mm (Width) × 630 mm (Height)

3.Legs for Cage Dimensions:
- 1120 mm (Length) × 850 mm (Width) × 900 mm (Height)

4.Cage Assembly Dimensions:
- 1100 mm (Length) × 800 mm (Width) × 700 mm (Height)

5.Gravity Conveyor Dimensions:
- 750 mm (Length) × 620 mm (Width) × 785 mm (Height)





FLOW CHARTS

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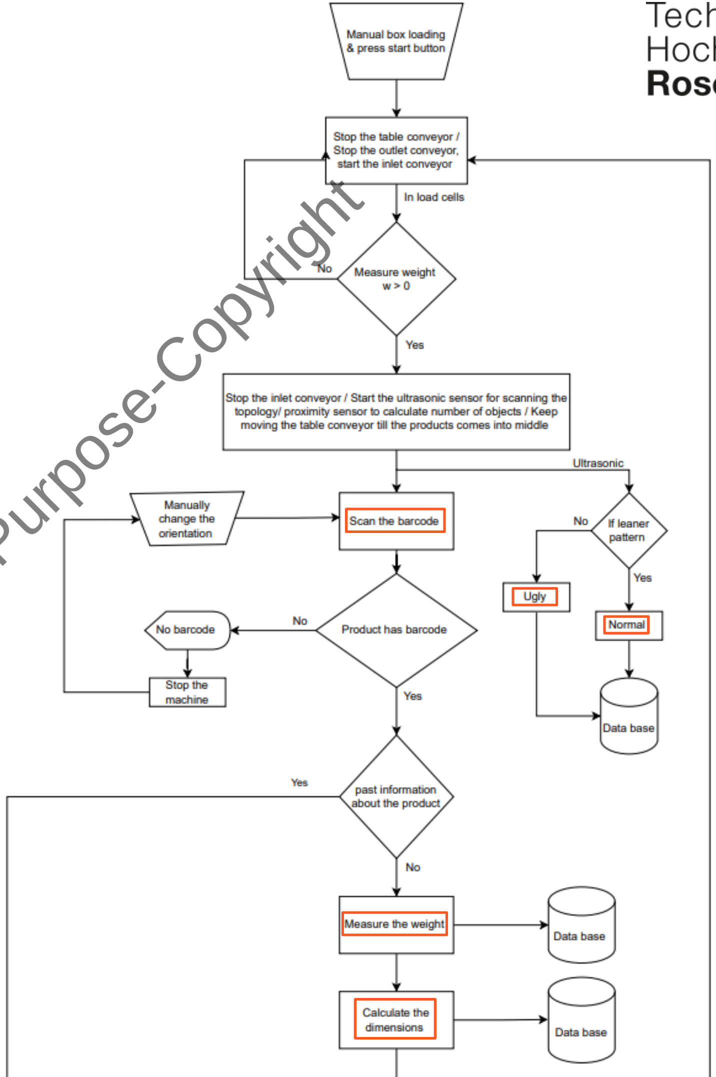
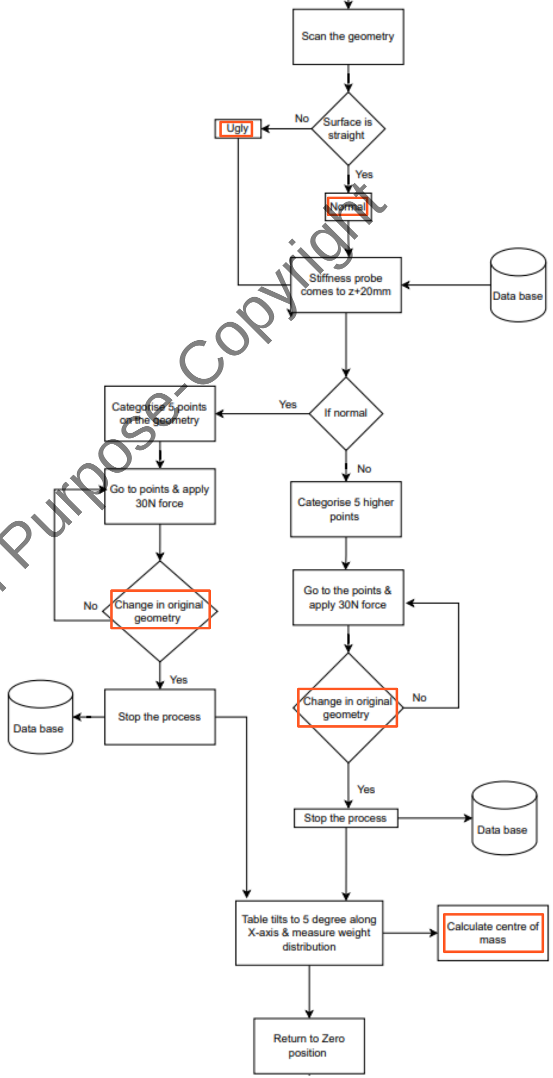
- **Start: Manual or Robotic Loading**
  - **Manual/Robotic Loading:** The product is loaded onto the machine manually by an operator or automatically by a robot.
  - **Machine Activation:** The operator turns on the machine.
- **Inlet Conveyor Activation**
  - **Inlet Conveyor Starts:** The conveyor begins moving the product towards the main table
- **Product Positioning and Measurement**
  - **Weight Measurement:** As the product moves to the main table, a load cell measures its weight.
  - **Conveyor Control:** The main table conveyor starts rotating, and the inlet conveyor stops.
  - **Surface Topology Detection:** Ultrasonic sensors detect the surface topology from the sides.
  - **Proximity Sensors:** Proximity sensors record the input time of the product.
- **Product Centering**
  - **3D Camera Feedback:** The product reaches the center of the table, stopping the conveyor based on feedback from the 3D camera



FLOW CHARTS

- **Barcode Scanning and Data Retrieval**
  - **Barcode Scanning:** The 3D camera scans the barcode to retrieve pre-installed data.
  - **Human Intervention:** If the barcode scan fails, human intervention corrects the product orientation.
  - **Data Check:** If the product is new, measurements are performed; otherwise, it passes through the system without measurements.
- **Product Measurement**
  - **Dimension Measurement:** The 3D camera measures the product dimensions.
  - **Weight Measurement:** The load cell provides the weight.
  - **Surface Topology Detection:** Ultrasonic sensors and the 3D camera analyze the surface topology to classify the product as "ugly" or "normal.“
- **Stiffness Measurement**
  - **Stiffness Probe:** A probe attached to a lead screw (for Z-axis movement) and a belt (for Y-axis movement) applies forces to determine product stiffness.



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FLOW CHARTS

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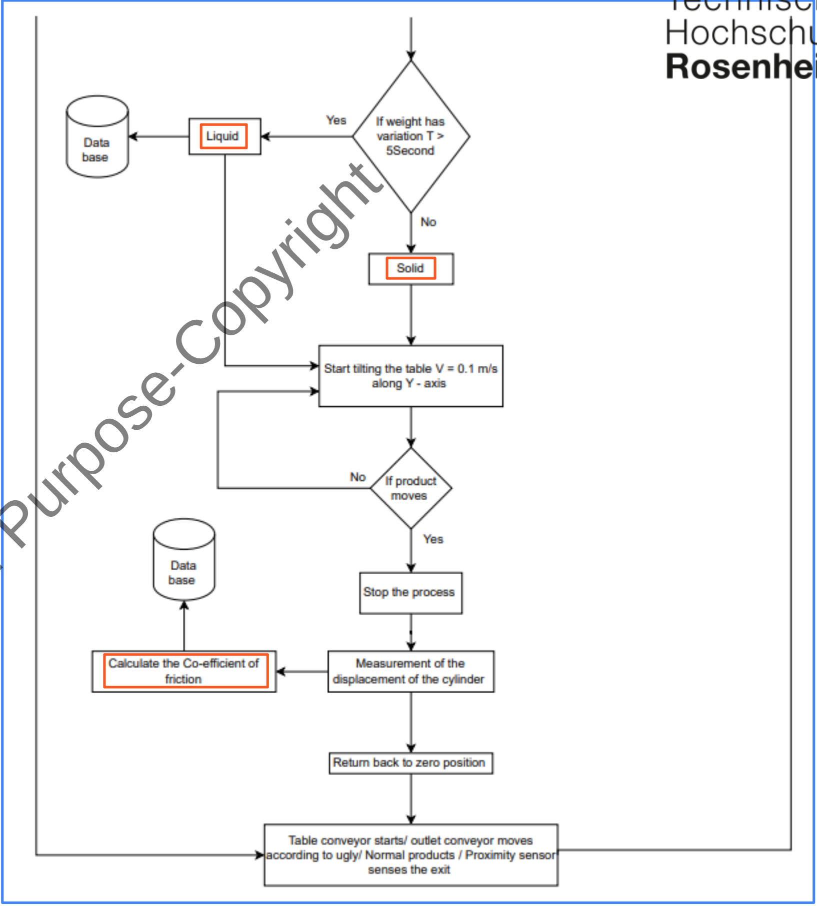
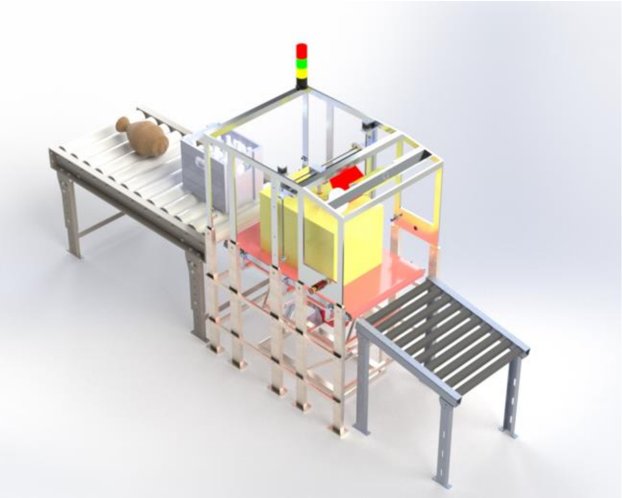
- **Center of Mass Calculation and Liquid/Solid Test**
  - **Table Tilting (X-axis):** The table tilts ±5 degrees to measure weight variation and determine if the product is liquid or solid.
  - **Returning to Zero Position:** After tilting, the table returns to its zero position.
- **Coefficient of Friction Calculation**
  - **Table Tilting (Y-axis):** The table tilts along the Y-axis until the product starts moving.
  - **Angle Measurement:** The angle at which the product starts moving is used to calculate the coefficient of friction.
- **Completion and Sorting**
  - **Return to Zero Position:** The table returns to its zero position after calculations.
  - **Conveyor Activation:** The table conveyor starts moving again.
  - **Outlet Conveyor Direction:** The outlet conveyor directs the product based on its classification ("ugly" or "normal").
- **End: Process Completion**
  - **Process Ends:** The product is sorted and the process completes.



- **This flowchart illustrates the step-by-step operation of the machine, from loading to the final sorting of the product.**

ESTIMATED TIME TAKEN



Teaching Station Estimated time for a Regular Product (in Seconds)

Total time with allowance | 63.38 Secs | 1 Min 4 Secs

Start	0															
Proximity sensor	0.002															
Ultrasonic sensor	0.24															
Forward movement		1														
3d camera (barcode scanning)			0.125													
3d camera (dimension)				0.5												
Forward movement(centre of station)					1											
Box																
Measurement at Corner																
X -Z movement						30										
Y movement							2									
Measurement at Centre																
X-Z movement								3								
Y movement									1							
Tool moving back to home										2						
Weight measuremnt											0.5					
Inclination(for liquid)												1				
waiting time													5			
Inclination for stiffness														3		
Proximity sensor															0.02	
Moving out																3	
End																Total Time	53.385

Teaching Station Estimated time for a UglyProduct (in Seconds)

Total time with allowance | 79.38 Secs | 1 Min 20 Secs

Start	0																					
Proximity sensor	0.002																					
Ultrasonic sensor	0.24																					
Forward movement		1																				
3d camera (barcode scanning)			0.125																			
3d camera (dimension)				0.5																		
Forward movement(centre of station)					1																	
Uglies																						
Measurement at high points																						
X -Z movement point 1						30																
Y movement point 1							2															
X -Z movement point 2								3														
Y movement point 2									2													
X -Z movement point 3										3												
Y movement point 3											2											
X -Z movement point 4												3										
Y movement point 4													2									
X -Z movement point 5														3								
Y movement point 5															2							
Tool moving back to home																2						
Weight measuremnt																	0.5					
Inclination(for liquid)																		1				
waiting time																			5			
Inclination for stiffness																				3		
Proximity sensor																					0.02	
Moving out																						3	
End																						Total Time	69.385

Fastest Time taken for Regular Product	53.38	
With +10 allowance ,Approximate time	63.38	
Fastest Time taken for Ugly Product:	69.38	
With +10 allowance ,Approximate time	79.38	
Avg time for a Product Box or Ugly	71.38	1 min 12 secs
No of Products per hour	50.43	**51**

OPERATORS ACTION

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Operator Actions in Machine Operation

1.Loading:

- **Manual Loading**: Operator manually loads boxes onto the Inlet conveyor
- **Robotic Loading**: Boxes are loaded onto the Inlet conveyor using a robotic system.

2.System Activation:

Operator starts the system, initiating the entire machine operation.

3.Manual Orientation Adjustment:

If the 3D scanner is unable to scan the barcode, the operator manually adjusts the orientation of the boxes.

4.Emergency Procedures:

In case of an emergency or malfunction, the operator presses the emergency stop button to ensure safety.

Note: A single qualified trainee is required for handling this teach-in station

COST ESTIMATION

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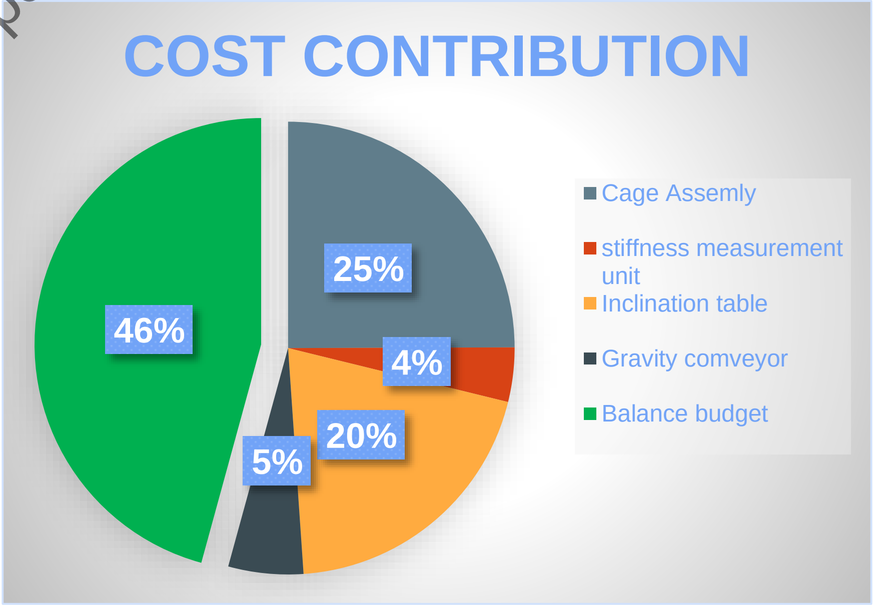
1. **Cage Assembly**
   - **Total Material Cost (First Prototype):** €3628.30
   - **Machining Cost (4 hours):** €400
   - **Operator Cost (4 hours):** €160
   - **Total Cost:** €4188.30
2. **Stiffness Measurement Mechanism**
   - **Total Material Cost (First Prototype):** €232.31
   - **Machining Cost (3 hours):** €300
   - **Operator Cost (3 hours):** €120
   - **Total Cost:** €652.31
3. **Inclination Table Assembly**
   - **Total Material Cost (First Prototype):** €2956.65
   - **Machining Cost (3 hours):** €300
   - **Operator Cost (3 hours):** €120
   - **Total Cost:** €3376.65
4. **Gravity Conveyor**
   - **Total Material Cost (First Prototype):** €481.50
   - **Machining Cost (3 hours):** €300
   - **Operator Cost (3 hours):** €120
   - **Total Cost:** €901.50

Component	Material Cost (€)	Machining Cost (€)	Operator Cost (€)	Total Cost (€)
Cage Assembly	3628.30	400	160	4188.30
Stiffness Measurement Mechanism	232.31	300	120	652.31
Inclination Table Assembly	2956.65	300	120	3376.65
Gravity Conveyor	481.50	300	120	901.50
Total				9188.76



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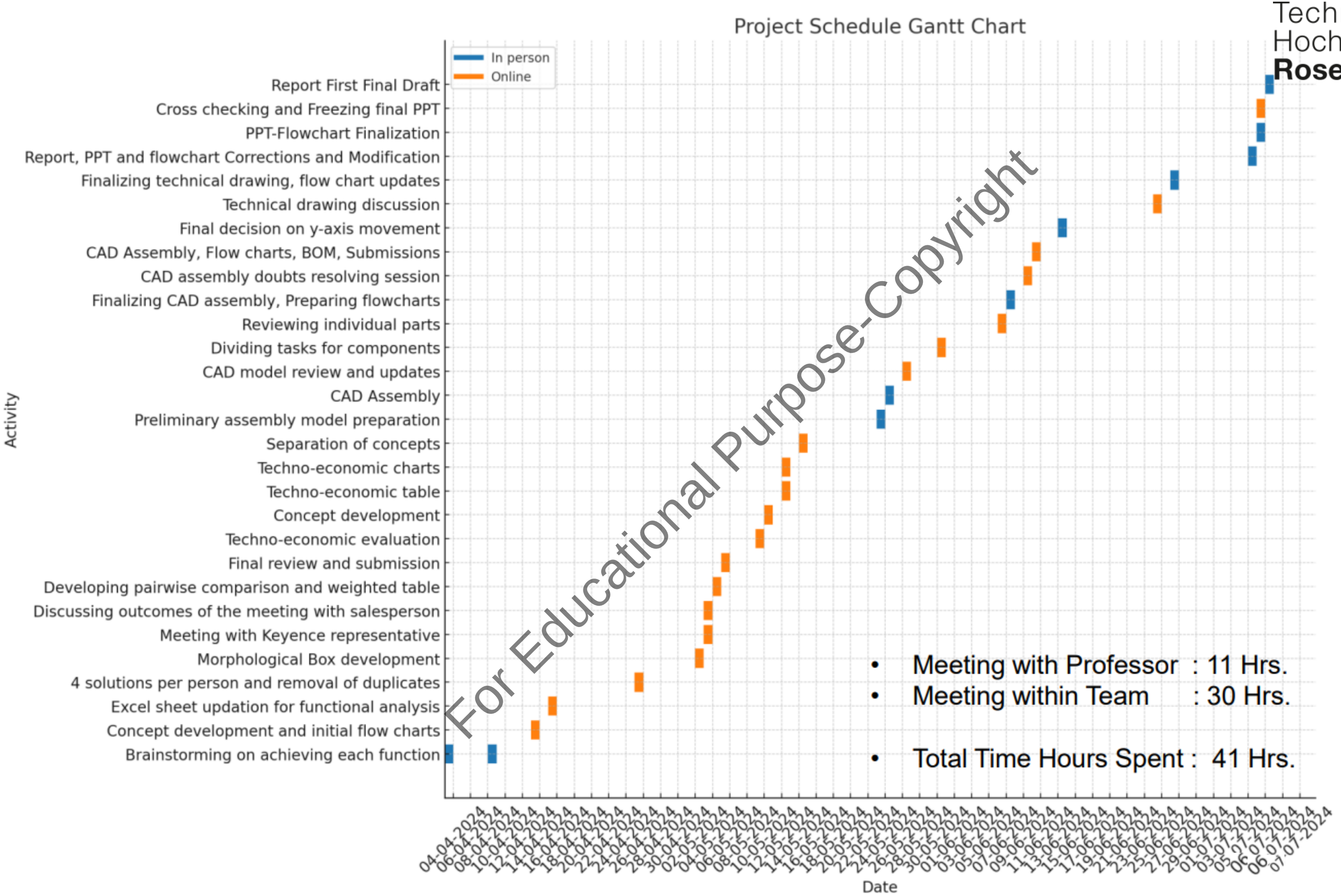

- Meeting with Professor : 11 Hrs.
- Meeting within Team : 30 Hrs.

- Total Time Hours Spent : 41 Hrs.

Thank You

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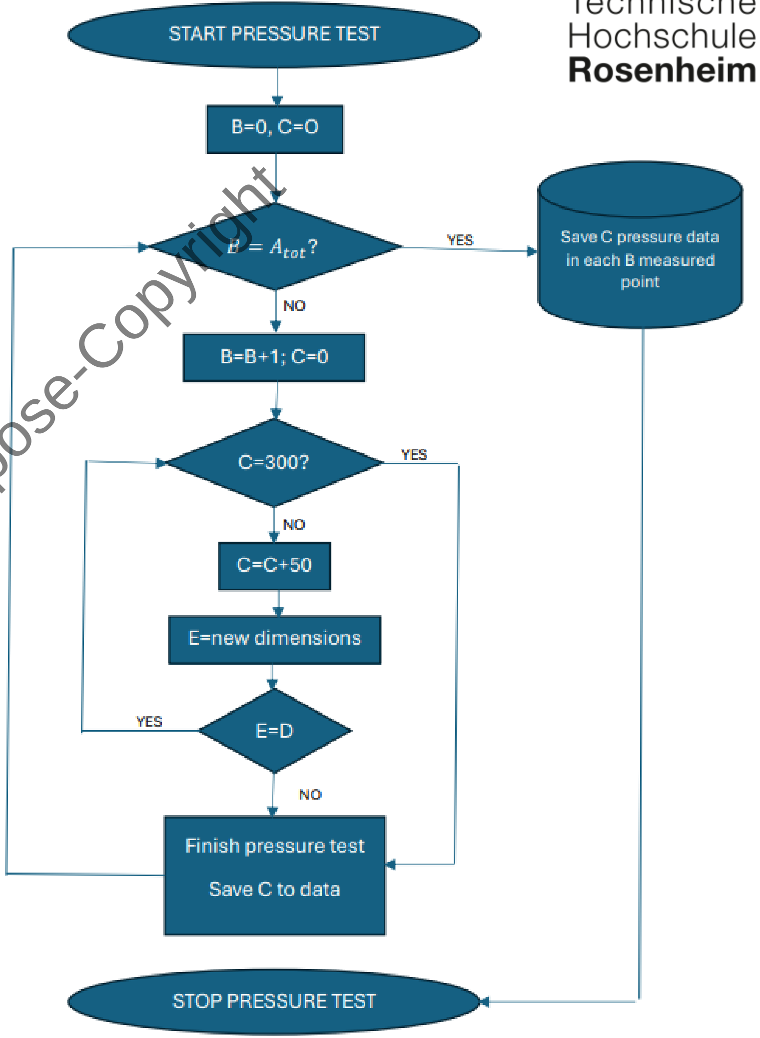
To measure the stiffness in P number of points the dimensions of the 3D camera supervise each range of force.

Variable	Description
A_{tot}	The total number of points is given by the software based on a ratio between the sizes and the final number of points, the variable assigned
B	Point for measure the stiffness
C	Force added
D	Data for dimensions
E	New compare dimensions



Determination of Centre of Mass

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1. **Record Load Cell Readings:**

   Let the readings from the four load cells be W_1, W_2, W_3, and W_4.

2. **Calculate Total Weight:**

   Sum the readings to get the total weight:

   $W_{total} = W_1 + W_2 + W_3 + W_4$

3. **Determine Load Cell Coordinates:**

   Assume the coordinates of the load cells are:

   - (x_1, y_1) for load cell 1
   - (x_2, y_2) for load cell 2
   - (x_3, y_3) for load cell 3
   - (x_4, y_4) for load cell 4

4. **Calculate Moments:**

   Compute the moments around the x and y axes:

   $M_x = W_1 \cdot x_1 + W_2 \cdot x_2 + W_3 \cdot x_3 + W_4 \cdot x_4$

   $M_y = W_1 \cdot y_1 + W_2 \cdot y_2 + W_3 \cdot y_3 + W_4 \cdot y_4$

5. **Calculate Center of Mass Coordinates:**

   The x and y coordinates of the center of mass are given by:

   $x_{CoM} = \frac{M_x}{W_{total}}$

   $y_{CoM} = \frac{M_y}{W_{total}}$

Example Calculation

Assume the load cells are positioned at the corners of a rectangular table with length L and breadth B:

- Load cell 1 at $(0, 0)$
- Load cell 2 at $(L, 0)$
- Load cell 3 at $(0, B)$
- Load cell 4 at (L, B)

Load Cell Readings:

- $W_1 = 10\,\mathrm{kg}$
- $W_2 = 15\,\mathrm{kg}$
- $W_3 = 20\,\mathrm{kg}$
- $W_4 = 25\,\mathrm{kg}$

Total Weight:

$W_{total} = 10 + 15 + 20 + 25 = 70\,\mathrm{kg}$

Moments:

$M_x = 10 \cdot 0 + 15 \cdot L + 20 \cdot 0 + 25 \cdot L = 40L$

$M_y = 10 \cdot 0 + 15 \cdot 0 + 20 \cdot B + 25 \cdot B = 45B$

Center of Mass Coordinates:

$x_{CoM} = \frac{40L}{70} = \frac{4L}{7}$

$y_{CoM} = \frac{45B}{70} = \frac{9B}{14}$

Determination of Centre of Mass –in 3D

1. **Use Load Cell Data for x and y CoM:**

   Use the previously described method to calculate the x and y coordinates of the CoM from the load cell readings.

2. **Use 3D Camera for z CoM:**

   Use the data from the 3D camera to estimate the height distribution of the object. The 3D camera can provide a detailed height map of the object, from which the vertical CoM can be estimated.

3. **Estimate Vertical Center of Mass:**

   - Capture a point cloud or height map of the object using the 3D camera.
   - Divide the object into small vertical segments or use the point cloud data directly.
   - Calculate the z-coordinate of the CoM by averaging the heights weighted by the local mass distribution, assuming uniform density or using known density variations.

Simplified Example:

Assume the 3D camera provides heights at a grid of points on the object's surface. For simplicity, consider a 2D grid of heights:

- Heights at grid points: $\{z_1, z_2, \ldots, z_n\}$
- Assume equal mass elements m (if density is uniform):

$$z_{CoM} = \frac{\sum_{i=1}^{n} z_i m}{\sum_{i=1}^{n} m} = \frac{\sum_{i=1}^{n} z_i}{n}$$

Integration with x and y CoM:

Combine the z-coordinate with the previously calculated x and y coordinates to get the full 3D center of mass $(x_{CoM}, y_{CoM}, z_{CoM})$.

Final Center of Mass:

Given the x, y, and z coordinates:

$$\mathrm{CoM} = (x_{CoM}, y_{CoM}, z_{CoM})$$